

Prefix Sum Queries

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to process q queries of the following types:

1. update the value at position k to u
2. what is the maximum prefix sum in range $[a, b]$?

Empty prefixes (with sum 0) are allowed.

Input

The first input line has two integers n and q : the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has three integers: either "1 k u " or "2 a b ".

Output

Print the result of each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $-10^9 \leq x_i, u \leq 10^9$
- $1 \leq k \leq n$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 4 1 2 -1 3 1 -5 1 4 2 2 6 1 4 -2 2 2 6 2 3 4	5 2 0